Электроника МКШ-2 — Display & Power Troubleshooting

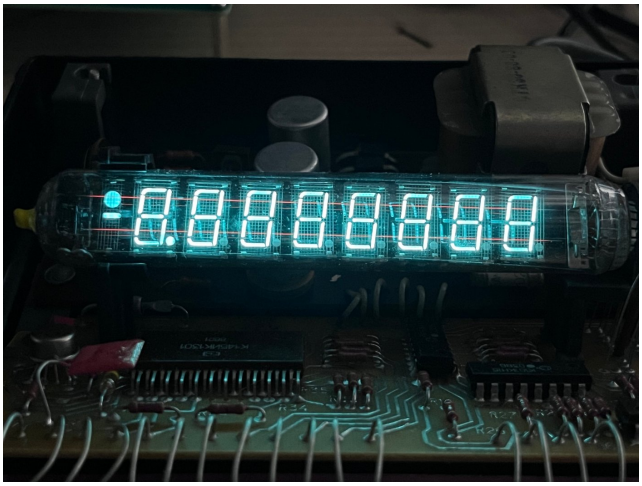
Workshop guide for the “all digits lit” fault and for setting up the 230V-to-42V converter. Part numbers and pin numbers come from the factory manual, appendix 1 (pages 29–30).

Unit / serial		Date	
Meter		Mains voltage at test	

> **Safety.** Only connect or move probes and clip leads with the converter **unplugged**. Inside the calculator the highest DC rail is about 45–50V. Mains voltage is present only inside the converter, so keep its enclosure closed. Plug in through an RCD-protected outlet. Hot resistors and filament wiring can burn.

1. Symptoms and what they mean

- The IV-18 is **multiplexed**. Its 17 lines are 9 grids (one per position) and 8 segment lines shared by every digit. Each grid switches on in turn while the segments for that digit are driven.
- A line is switched **on** (to 0V) by a К161КН1Б driver and pulled back **off** to Г (−45V) by its own 56kΩ resistor (R1–R17). The filament sits a few volts above Г (set by R38/VD2), so a line at Г is cut off.
- A blank digit is made by driving **no** segments during its time slot. If lines are not pulled back to Г, the display shows stale data.
- If digit 1 shows a sensible “0.” and the display reacts to keys, the processor (К145ИК1301) and keyboard are working.



Pattern A: every digit shows the same “0.” at full brightness, with all decimal points lit. Red filament wires mean the supply is too high.



Pattern B: digit 1 correct; blank digits show copies of a real digit that fade away the further they are from it (here after typing 456).

Display pattern	Most likely cause	Go to
A: same character in every digit, full brightness	Grids not turning off: filament bias missing, or grid pull-downs not reaching Г	Steps 5, 6
B: blank digits show fading copies of a real digit	Segment lines not pulled back to Г: broken common return, or leakage	Step 6 (6b, 6d), step 9 cleaning
Faint “ghost” segments behind real digits	Bias slightly low, supply too high, or board leakage	Steps 3, 5, 9
One segment or one digit always on or always off	One pull-down, one driver output, or one display lead	Step 6 (6a, 6h, 6i)

Display pattern	Most likely cause	Go to
Pattern changes when the board is handled	Dry joint or cracked track	Step 6 (6e)
Visible red/orange filament glow	Supply too high	Steps 3, 8

2. Quick check: clear the calculator

☐ Switch on and press **Cx**. According to manual section 7.3, the calculator is ready when the display shows only “0.”.

☐ If anything else is displayed, switch off, wait **at least 3 seconds**, and switch on again (sections 7.3, 8.1.2).

☐ Repeat with a few entries (for example ВП, then 4 5 6) and photograph the display each time. If digits that should be blank are lit, match the pattern to the table in step 1 and continue below.

3. Supply voltage at the calculator input

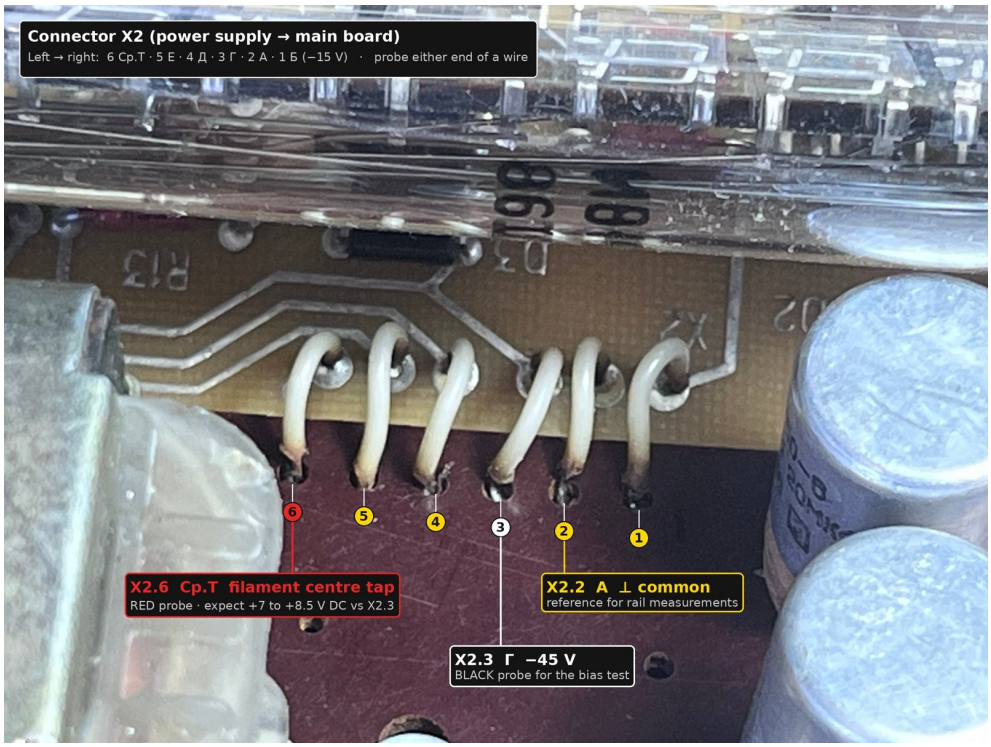
Specification (manual 3.10): **42V AC +10% / −15% at 50 ± 1Hz**, which allows **35.7–46.2V**. Power consumption is 9W at most (3.7), which is about 0.2A.

Measurement	Expected	Measured
Converter output, calculator connected and on (V AC)	42V (35.7–46.2V)	
Calculator input current (A AC, meter in series with one leg)	≤ 0.21A	

4. Internal supply rails at connector X2

Calculator running. Use the DC range unless marked AC, and measure relative to **X2 pin 2 (common, “+Общий”)** unless stated otherwise. The −15V rail is regulated. The −45V rail is **not** regulated and rises in proportion to the input voltage.

X2 pin	Net (schematic)	Expected	Measured
1	Б — −15V logic supply	≈ −15V DC (steady)	
2	А — common ⊥ (+Общий)	0V, reference	—
3	Г — −45V display drive	≈ −45V at 42V in; ≈ −50V at 48V in	
4 ↔ 5	Д / Е — IV-18 filament (Накал)	≈ 5V AC between pins 4 and 5	
6	Ср.Т — filament centre tap	See step 5	



Connector X2 on the main board, pins numbered from the right: 6 Ср.Т · 5 Е · 4 Д · 3 Г · 2 А · 1 Б. Each wire is one pin, so probe either end.

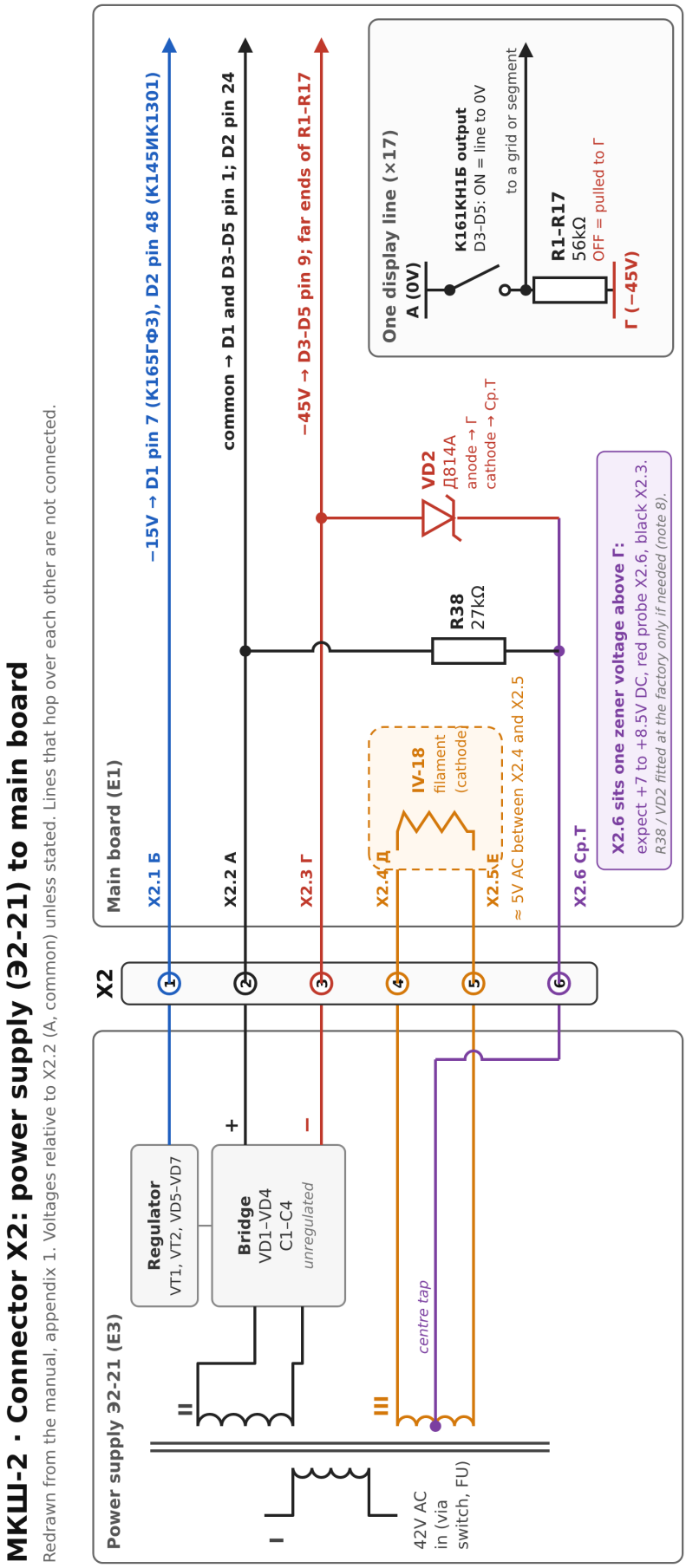
5. The key test: filament bias

The display's 56kΩ pull-down resistors hold the "off" grids and segments at the −45V rail (Г). For them to stay dark, the filament must sit a few volts **above** −45V. R38 and VD2 set that level on the centre tap (X2 pin 6).

> **Measure (DC):** black probe on **X2 pin 3 (−45V)**, red probe on **X2 pin 6 (centre tap)**.
> **Expect: about +7 to +8.5V** (the Д814А zener voltage).
> Measured: _____ V Polarity confirmed (+ on X2.6)? ☐ Yes ☐ No

Reading (pin 6 relative to pin 3)	Likely cause	Action
About +7 to +8.5V	Bias is correct	Go to step 6 (pull-downs and drivers)
About 0V, or negative	VD2 shorted, or centre tap tied to −45V	Replace VD2 (step 7)
Low or drifting	R38 open (zener gets no current)	Replace R38 (step 7)
R38 / VD2 not on the board	Not fitted at the factory (they were fitted only when needed)	Fit both (step 7)

The X2 interface, including R38 and VD2, is drawn in full on the next page (turn the page sideways).



Connector X2, redrawn from the manual (appendix 1, pages 29–30).

R38 (27kΩ) runs from X2.6 to common (A). VD2 (Д814А) has its **anode on Г (X2.3)** and its **cathode on X2.6**, so the filament centre tap sits one zener voltage above Г.

Note 8 of the schematic (translated): “Fitted during adjustment at the factory if unused digits of the display glow.”

The inset shows how each of the 17 display lines is switched on by a К161КН1Б output and pulled off to Г by its 56kΩ resistor.

6. Pull-downs and display drivers (if the bias is correct)

R1–R17 (56kΩ, next to the display) are one pull-down per display line: **8 for the segment lines** and **9 for the grids**. Their far ends join on a common track back to Г (X2.3). The two groups may share one track or use two. **Measuring across each resistor proves the resistor, not its return:** every resistor can read 56kΩ while the shared end is disconnected from Г, in which case none of them pulls anything down.

Lines	IV-18 pins (from the manual's character table)	Fault if not pulled down
Segments (8)	2 = decimal point, 9 = middle bar / minus, 3, 4, 5, 10, 11, 12 = the other six segments	Pattern B: blank digits copy a real digit, fading with distance
Grids (9)	G1–G9, one per position (service digit + 8 digits)	Pattern A: every digit shows the same character

#	Test	Expected	Measured
6a	**Power off.** Resistance across each of R1–R17, in circuit	≈ 53–58kΩ (a little under 56kΩ from parallel paths)	Lowest ____ Highest ____
6b	**Power off.** Common end of the **segment** pull-downs to X2.3 (Г)	≈ 0Ω	
6c	**Power off.** Common end of the **grid** pull-downs to X2.3 (Г)	≈ 0Ω	
6d	**Clip-lead test.** Power off, clip a lead from the common end that failed 6b/6c to X2.3, then power on	Blank digits go dark = fault confirmed	☐ Clears ☐ No change
6e	**Flex / tap test.** Meter on 6b/6c while flexing the board gently; then power on and tap around R1–R17, D3–D5 and the IV-18 leads with an insulated stick	Steady reading; display does not change	Location: _____
6f	**Power on.** К161КН1Б D3, D4, D5 pin 9 relative to X2.2	≈ −45 to −49V on each	D3 ____ D4 ____ D5 ____
6g	**Power on.** К161КН1Б D3, D4, D5 pin 1 relative to X2.2	0V on each	D3 ____ D4 ____ D5 ____
6h	**Power on, DC.** A **segment** line at the display end of its pull-down, relative to X2.2	Strongly negative (tens of volts); near 0V = floating	
6i	**Power on, DC.** A **grid** line at the display end of its pull-down, relative to X2.2	Strongly negative (≈ −40V average); near 0V = floating	
6j	**Scope (optional).** Segment and grid lines, compared with manual trace 5 (Д4 with "0" displayed)	Sharp pulses to 0V, straight back to ≈ −45V; no slow decay	

- ☐ 6b or 6c open or high: repair the break in that common return (reflow joints, or bridge the track with a short wire). Then repeat 6d without the clip lead.
- ☐ A resistor out of range in 6a: replace it, then check whether its line now switches off.
- ☐ 6e finds a sensitive spot: reflow every joint there under a magnifier, including the driver pins and the IV-18 leads.
- ☐ 6f missing −45V on a pin 9: trace the Г feed to that driver.
- ☐ 6h or 6i near 0V with 6b/6c passing: suspect leakage (step 9, clean the board) or a driver output stuck on.

7. Fit or replace R38 and VD2 (only if step 5 failed)

Ref	Original	Modern replacement	Notes
R38	27kΩ МЛТ-0.125	27kΩ 0.25W metal film	Value chosen at the factory (**).
VD2	Д814А zener, about 7–8.5V	8.2V 0.5–1W zener: 1N4738A or BZX79-C8V2	If faint glow remains, try 9.1V, then 10V.

☐ Unplug. Fit VD2 with its **anode (unbanded end) to X2.3 (Г)** and its **cathode (banded end) to X2.6 (Ср.Т)**. Fit R38 between X2.6 and common (A).

☐ Power up and repeat the step 5 measurement. Expect about the zener voltage.
Measured after fitting: _____ V

☐ Display shows a single “0.” and the other digits are dark.

☐ Don’t use more bias than you need: a higher zener also dims the segments that should be lit.

8. Series dropper resistor and output fuse (bring the supply down to 42V)

A converter that delivers more than 42V under load overdrives the IV-18 filament (visible red glow) and raises the −45V rail. A series resistor brings both back into specification. On its own it does not cure the display fault.

- **Series only.** Use one resistor in either output leg, or split the value across both legs (still in series). Never wire them in parallel: 22Ω ‖ 27Ω is only about 12Ω.
- Fit it **inside the converter**, not inside the calculator. Order on the converter output: **transformer → fuse → resistor → socket**.
- Choose the rating from **P = I² × R**, and use a resistor rated for at least 2× the calculated power, preferably 3× or more.
- **The calculator’s current depends on the display fault.** Lit segments draw current, so the current can fall noticeably once the display is fixed. Size the final dropper after the repair, measuring with the dropper fitted.

Method

☐ Calculator running, dropper fitted. Measure the AC voltage **across each resistor**: I = V ÷ R. Both should give the same current (this also confirms a series connection).

☐ Measure the transformer output **before** the dropper (V_T) and the voltage **at the calculator**, after the dropper.

☐ Required total: R = (V_T − 42) ÷ I. Refit, then measure again at the calculator, and check the filament (X2.4–X2.5 ≈ 5V AC).

Option (example: V_T ≈ 50V, I ≈ 81mA)	Total	Power	Expected at calculator	Notes
100Ω 3W wirewound or metal oxide	100Ω	≈ 0.65W	≈ 42–42.7V	Single part
2 × 51Ω 3W (e.g. TE/Neohm 5-2176793-0), one per leg	102Ω	≈ 0.33W each	≈ 42–42.7V	Same layout as a two-leg test rig
33Ω + 68Ω, each ≥ 1W	101Ω	≈ 0.22W + 0.45W	≈ 42.7V	From common values

Fuse	Rating	Why
Output	**About 2× the running current, slow-blow, 250V**, 5×20mm, same leg as the resistor, before it. Example: 160mA T with a ≈ 100Ω dropper	A short is limited by the dropper (≈ V_T ÷ R, ≈ 0.5A here). The fuse must be well under a third of that, so it blows in seconds before the resistor overheats.
Input (existing)	200mA slow-blow (230V); about 400mA slow-blow for a 115V build	Mains-side faults only. It cannot see an output short.

R = (V_T − 42) ÷ I **P** = I^2 × R Example: V_T = 50V, I = 0.081A → R ≈ 100Ω, P ≈ 0.65W

V_T = _____ V V across each resistor: _____ / _____ V

I = _____ A R = _____ Ω Fitted: _____ Ω / ____ W Fuse: _____ mA

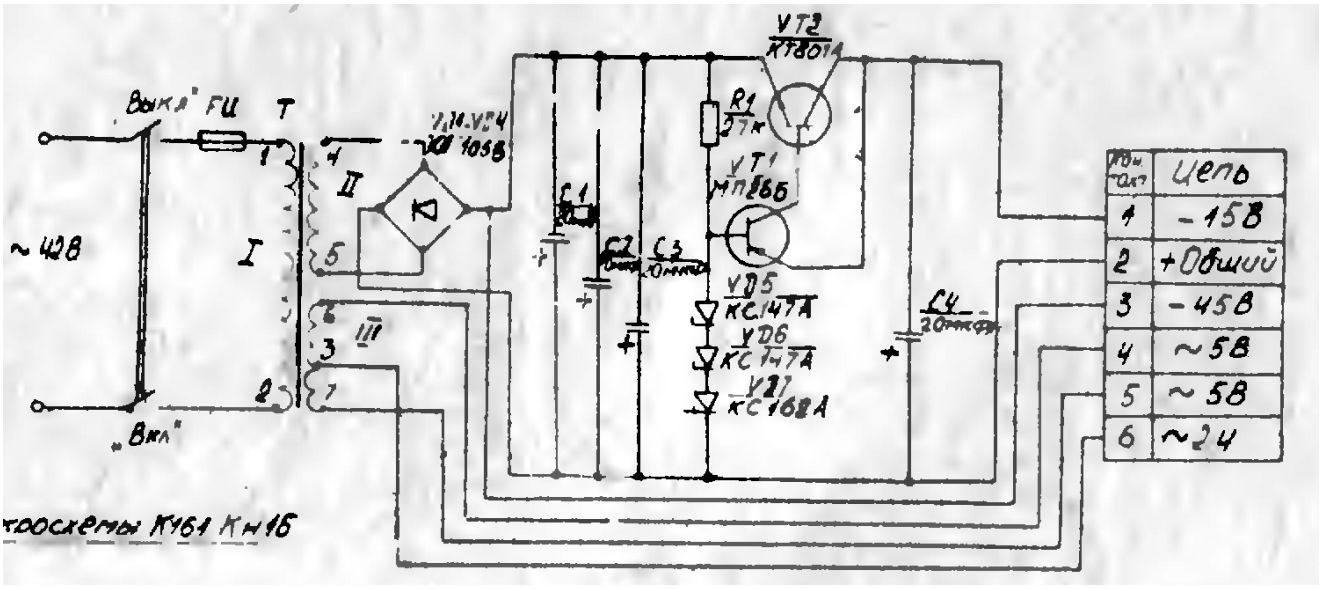
Voltage at calculator after fitting: _____ V Filament X2.4–X2.5: _____ V AC (target ≈ 5V)

Target: about 42V at the calculator with it running. The output still follows the mains, so ±1–2V through the day is normal and within specification. The filaments should no longer glow visibly red.

9. If the display is still wrong

- ☐ **Clean the main board.** Old flux, dust and moisture between the closely spaced −45V tracks can hold display lines partly on. Unplug, then scrub with isopropyl alcohol (99%) and a soft brush around D3–D5, R1–R17, the IV-18 leads and X2. Let it dry completely (at least an hour) before powering up. Note the display pattern before and after.
- ☐ **Electrolytics C1–C4** (K50-6, 100V) in the power supply block: check ESR, or measure AC ripple on the −15V and −45V rails. More than a few hundred mV is suspect. Ripple on −45V weakens the cutoff.
- ☐ **К161КН1Б drivers D3–D5:** with the step 6 tests passing, an output stuck on points to a failed driver.
- ☐ **Solder joints:** check the IV-18 wire leads, the driver chips and the X2 connector under a magnifier, and reflow anything doubtful. Also check the insulation of the X2 wires for heat damage or cracks where they are soldered.
- ☐ Final check: run the **74-step test** in manual Table 3 (pages 11–13) and compare the display after each key.

Reference: power supply block Э2-21 (manual page 30)



Ref	Part	Function
T, FU	Transformer, fuse	42V AC in. Winding II feeds the bridge; winding III is the filament winding with centre tap.
VD1–VD4	КД105В	Bridge rectifier, giving the unregulated −45V rail
C1–C4	К50-6, 100V	Filter and reservoir electrolytics. Suspect them first in a unit this old.
VT1, VT2	МП26Б, КТ801А	Series regulator for the −15V rail
R1	27kΩ	Regulator bias
VD5, VD6, VD7	КС147А, КС147А, КС168А	Reference chain for −15V (about 4.7 + 4.7 + 6.8V). The manual lists alternatives (Д814А, КС156А, etc.).

Item	Value
IV-18 filament	5V, 0.085A, directly heated
Display drive	On = common (0V) through К161КН1Б; off = −45V through 56kΩ pull-downs
Display lines	9 grids (G1–G9) + 8 segments (pins 2, 3, 4, 5, 9, 10, 11, 12) = R1–R17
Filament bias	X2.6 ≈ +7 to +8.5V relative to X2.3 (R38 27kΩ to common; VD2 Д814А, anode to X2.3, cathode to X2.6)
Logic	К145ИК1301 processor and К165ГФ3 clock generator on −15V
Supply input	42V AC +10% / −15%, 50 ± 1Hz, ≤ 9W

Compiled from the Электроника МКШ-2 operating manual (Svetlovodsk, appendix 1) and bench observations. The schematic scans are low quality, so check part values and orientations against the board before replacing anything.